NATIONAL UNIVERSITY OF SINGAPORE
SCHOOL OF COMPUTING
MID-TERM TEST FOR
CS3244: MACHINE LEARNING

This is a sample set of answers for the midterm exam.

The email disclosing your answers may be for a different ordering of both the questions and answer options; as we needed to create versions of the test to allow you all to take the exam next to each other in the LT 19 venue without worrying about copying.

Thus the correct answer options indicated on your email may differ from those indicated correct in this paper.

Multiple Choice Questions are scored in a binary (full marks / no marks) fashion. For Q23, there were multiple correct choices, as the answer fell on a border; so both have full marks.

Multiple Response Questions are fractionally scored as the number of correct values per binary letter choice, corrected for random chance guessing.

That is, we view each of the MRQs as 5 binary questions (e.g., Yes "A" or No "A"). But by random chance, we would expect to get 2.5 of these 5 questions correct. We thus need to correct for random guessing. We do this by subtracting 2 (rounding down from the appropriate value of 2.5) from your total of binary questions that you got right. Thus the maximum value would be 3 of 3 binary questions, for full marks for that question and either a 2/3rds or 1/3rd partial marks for other cases.

E.g., if the response key is "AB" and you answered "ABC", you would have gotten "A, B, no D, no E" correct and the binary question for C wrong (4 of 5 binary questions on each of the letters correct); 4 correct responses. After the random correction of subtracting 2 questions, we obtain 4-2 = 2 of the possible 3 binary questions correct.

In the assessment, each of the MRQs was worth 2 marks.

Here is a worked examples of the scoring (assuming a correct answer is "AB" worth 2 marks):

Sample Response	# of 5 binary questions correct	After –2 random guess correction	Marks for the question (given MRQ is worth 2 marks)
CDE	0	0 of 3 possible = 0	0
CD	1 (no E)	0 of 3 possible = 0	0
C	2 (no D, no E)	0 of 3 possible = 0	0
AC	3 (A, no D, no E)	1 of 3 possible = 1/3	0.66
ABC	4 (A, B, no D, no E)	2 of 3 possible = 2/3	1.33
AB	5 (A, B, no C, no D, no E)	3 of 3 possible = 3/3	2.0

Section 1: Paradigms of Machine Learning.

Q2: Which of the below are examples of the *Reinforcement Learning* paradigm? Mark all responses that apply. If none fit, do not mark any.
Answers: A, C

Q3: Which of the below are examples of the *Unsupervised Learning* paradigm? Mark all responses that apply. If none fit, do not mark any.
Answer: D

A. Robot navigating to exit from a crowded plane during an emergency.
B. In a game of soccer, identifying who is a striker and who is a defender. You are given the following player data from last 10 matches: Goals scored, goals saved, number of assists, number of yellow cards, number of free-kicks, number of successful tackles.
C. Designing a traffic light controller to solve the congestion problem in a busy area with multiple traffic signals.
D. Segmenting customers based on customer data, purchase history, behavior, frequency of visit etc.
E. From the perspective of a self-driving car, identifying people, cars and trees from images.

Section 2: Concept Learning

Q4: For a hypothesis h that is a conjunction of constraints on some input attributes, the number of possible instances is $|X|$, the number of possible hypotheses is $|H|$.

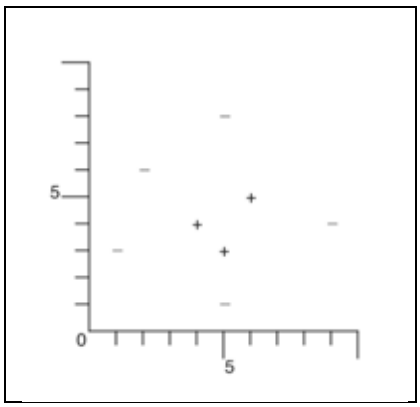
A new input attribute that can take on k values is introduced. Calculate the new number of possible input instances, and number of semantically distinct hypotheses, respectively. Mark one choice.

A. $(k+1)|X|$, $|H|(k+1)-k$
B. $k|X|$, $|H|(k+1)-k$
C. $k|X|$, $|H|(k+1)$
D. $(k+1)|X|$, $|H|(k+1)$
E. None of the above.

Adding a new input attribute with k possible values yields $k|X|$ possible instances and $(1+(k+1)(|H|-1)) = |H|(k+1)-k$ possible hypotheses.

Q5: Consider the instance space consisting of integer points in the x, y plane and the set of hypotheses H consisting of rectangles (*i.e.*, hypotheses are of the form $a \le x \le b, c \le y \le d$, where a, b, c, and d are integers). Consider the version space with respect to the set of positive (+) and negative (–) training examples shown below.

What are the correct S and G boundaries that the define the version space? Mark one choice.



A. $S = \{(4 \le x \le 6, 4 \le y \le 5), (4 \le x \le 5, 3 \le y \le 4)\}$,
$G = \{(2 \le x \le 8, 2 \le y \le 5), (3 \le x \le 8, 2 \le y \le 7)\}$.
B. $S = \{(4 \le x \le 6, 4 \le y \le 5), (4 \le x \le 5, 3 \le y \le 4)\}$,
$G = \{(1 \le x \le 9, 3 \le y \le 6), (2 \le x \le 5, 1 \le y \le 8)\}$.
C. $S = \{(4 \le x \le 6, 3 \le y \le 5)\}$,
$G = \{(2 \le x \le 8, 2 \le y \le 5), (3 \le x \le 8, 2 \le y \le 7)\}$.
D. $S = \{(4 \le x \le 6, 3 \le y \le 5)\}$,
$G = \{(1 \le x \le 9, 3 \le y \le 6), (2 \le x \le 5, 1 \le y \le 8)\}$.
E. None of the above.

The gist is that the S boundary needs to be as tight as possible but enclose all positive examples, and that the G boundar(ies) be as loose as possible, enclosing the positives, without including any negatives.

Q6–7: Consider the below scenario for a friend's preference for components of a healthy meal, with the following attribute values for each component: $X1$ = {White_rice, Brown_rice}, $X2$ = {Cauliflower, Broccoli}, $X3$ = {Carrot, Radish}, $X4$ = {Ice_water, Warm_water}.

Now let's say that we have partially run conjunctive concept learning using the *Candidate–Elimination* algorithm and currently have a version space defined by:

G = ⟨White_rice, ?, ?, ? ⟩; ⟨?, ?, ?, Warm_water⟩; ⟨?, ?, Carrot, ? ⟩
S = ⟨White_rice, ?, Carrot, Warm_water⟩

And we have the additional training instances that we need to consider:

$x^{(1)}$: ⟨White_Rice, Broccoli, Carrot, Ice_water⟩, *positive*.
$x^{(2)}$: ⟨Brown_Rice, Cauliflower, Radish, Ice_water⟩, *negative*.

Q6 (4 points): Which of the following hypotheses will be in the version space after running *Candidate–Elimination* on the above examples $x^{(1)}$ & $x^{(2)}$, starting from the G and S boundaries, where applicable? Mark all responses that apply. If none fit, do not mark any.

A. ⟨White_rice, ?, Carrot, ? ⟩.
B. ⟨White_rice, ?, ?, Warm_water⟩.
C. ⟨White_rice, ?, Carrot, Warm_water⟩.
D. ⟨Brown_rice, ?, Carrot, ? ⟩.
E. ⟨?, ?, Carrot, Warm_water⟩.

Running the Candidate-Elimination algorithm, will eliminate the inconsistent hypothesis for both positive and negative examples. If we start from current S and G and run through the first example the attribute Warm_water is no longer consistent, so we remove it from our S. The updated S will be: S = ⟨White_rice, ?, Carrot, ?⟩. For G, we remove the second hypothesis ⟨?, ?, ?, Warm_water⟩, as it is no longer consistent. For the negative example, the updated hypotheses are the same as previous step. So finally, our S and G are, S = ⟨White_rice, ?, Carrot, ?⟩ and G = ⟨White_rice, ?, ?, ?⟩; ⟨ ?, ?, Carrot, ?⟩

Q7: Which of the following hypothesis is consistent after running *Find–S* on the above examples $x^{(1)}$ & $x^{(2)}$, starting from the G and S boundaries, where applicable? Mark all responses that apply. If none fit, do not mark any.

A. ⟨White_rice, ?, ?, ? ⟩.
B. ⟨?, ?, Carrot, ? ⟩.
C. ⟨White_rice, ?, Carrot, ? ⟩.
D. ⟨White_rice, ?, Carrot, Warm_water⟩.
E. *Find–S* returns only a specific hypothesis.

Recall that *Find-S* returns only a specific hypothesis and only updates with positive example. So we start with the current S, which is S = ⟨White_rice, ?, Carrot, Warm_water⟩. If we evaluate the positive example then the attribute Warm_water is not consistent any more. So our specific hypothesis will update to S = ⟨White_rice, ?, Carrot, ?⟩.

Section 3: Naïve Bayes

Consider the problem of binary classification using the Naïve Bayes classifier, where you are given two categorical features x_1 and x_2 and their categorical class conditional distributions in the tables below.

The entries in the tables correspond to $P(X_1 = x_1|C_i)$ and $P(X_2 = x_2|C_i)$, respectively. Note that the two classes are equally likely *a priori*.

	Class	
X_1	C_1	C_2
-1	0.2	0.3
0	0.4	0.6
1	0.4	0.1

	Class	
X_2	C_1	C_2
-1	0.4	0.1
0	0.5	0.3
1	0.1	0.6

Q8 (<u>4 points</u>): Which of the following probability values are correct? Mark all responses that apply. If none fit, do not mark any.

A. $P(X_1 = 1, X_2 = -1) = 3/5$.
B. $P(C_1 | X_1 = 0, X_2 = -1) = 8/11$.
C. $P(C_1 | X_1 = -1, X_2 = 1) = 1/5$.
D. $P(C_2 | X_1 = 0, X_2 = -1) = 3/11$.
E. $P(C_2 | X_1 = -1, X_2 = 1) = 4/5$.

Use Bayes' Rule and the Naïve Bayes assumption to solve for the probabilities. Example:

$$P(C_1 \mid X_1 = 0, X_2 = -1) = P(X_1 = 0, X_2 = -1 \mid C_1) \times P(C_1) \,/\, P(X_1 = 0, X_2 = -1)$$
$$= P(X_1 = 0 \mid C_1) \times P(X_2 = -1 \mid C_1) \times P(C_1) \,/\, P(X_1 = 0, X_2 = -1)$$

$$P(X_1 = 0, X_2 = -1) = P(X_1 = 0 \mid C_1) \times P(X_2 = -1 \mid C_1) \times P(C_1) + P(X_1 = 0 \mid C_2) \times P(X_2 = -1 \mid C_2) \times P(C_2)$$

<u>Q9–10</u>. The Naïve Bayes algorithm you have seen in lecture can only be applied to discrete feature space. In this question, let's extend it to handle continuous feature spaces.

Here's the idea: Instead of estimating $P(x_i | C_k)$ with a categorical distribution for each x_i and $C_k = y$, we use a normal distribution, parameterized by a mean μ_k and variance σ_k^2. These parameters are estimated with the empirical mean and variance of the data, respectively.

Q9: Suppose we have m training examples, n features, and k classes. How many model parameters do we need? Mark one choice.

A. nk
B. $2nk$
C. $2nmk$
D. $k \times 2^n$
E. $2k \times 2^n$

We need one pair of parameters to model mean and variance for each combination of class k and parameter n. The number of parameters is not sensitive to the number of training instances, as with standard Naïve Bayes.

Q10 (<u>8 points</u>): Given the data in the table on the right, compute the conditional probability $P(x_2 = 1|y = 1)$. Mark one choice.

N.B.: $P(x = v|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(v-\mu_k)^2}{2\sigma_k^2}}$ and $\sigma = \sqrt{\frac{\sum(x-\mu)^2}{n-1}}$.

x_1	x_2	y
2.9	3.2	0
3.3	2.8	0
1.0	4.1	0
2.4	1.5	1
2.2	–1.5	1

A. $P < 0.00001$

B. $0.00001 \leq P < 0.05$
C. $0.05 \leq P < 0.2$
D. $0.2 \leq P < 0.25$
E. $P \geq 0.25$

The values of x_2 that have $y = 1$ are 1.5 and -1.5. Their mean is 0, and their variance is 4.5. By substituting these values into the formula for normal distribution, we get:

$$(1/\sqrt{(2\pi * 4.5)})\, exp(-1/(2 * 4.5)) \;=\; 0.1683.$$

Section 4: Linear Models

Q11: Let's consider a variant of logistic regression that uses the hyperbolic tangent function instead, $tanh\ (s) \;= \frac{e^{2s}-1}{e^{2s}+1}$. Which of the following are true statements about $tanh\ (s)$? Note that $tanh\ (x) \;= 2\sigma(2x) - 1$, where σ is the logistic sigmoid function $\sigma(x) = \frac{e^x}{1+e^x}$. Mark all responses that apply. If none fit, do not mark any.

A. Its range is $(-1,1)$.
B. This function is strictly increasing.
C. The function is *convex*; *i.e.* for any point $x_1, x_2 \in R$ and $\alpha \in [0,1]$, $f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2)$.
D. The function is *concave*; *i.e.* for any points $x_1, x_2 \in R$ and $\alpha \in [0,1]$, $f(\alpha x_1 + (1-\alpha)x_2) \geq \alpha f(x_1) + (1-\alpha)f(x_2)$.
E. $\tanh(x)$ horizontally compresses and vertically stretches (rescales) the logistic sigmoid function.

The function is strictly increasing since sigmoid is strictly increasing. Its range is $(-1, 1)$ because the sigmoid takes on $(0,1)$, and after scaling and shifting, we obtain $(-1,1)$. Function is neither convex nor concave. tanh also both horizontally stretches (not compresses) and vertically stretches the standard logistic function: $tanh(x) \;=\; 2\sigma\ (2x) \;-\; 1$.

12: Consider a perceptron with two inputs, $x_1$and x_2. Supposing the weights θ_1 and θ_2 of this perceptron are both set to the value of 0.5, what is the largest possible range of the values of the bias weight, θ_0 that can be set for the perceptron to represent the AND gate – *i.e.*, AND(x_1, x_2).

x_1	x_2	(x_1, x_2)
		–1
		–1
		–1
		1

Assume that the inputs x_1 and x_2 and output $o(x_1, x_2)$ of the perceptron are Boolean with the values of 1 or –1, and an output $o(x_1, x_2) \;=\; 0$ is considered to be negative. Mark one choice.

A. $-1 < \theta_0 \leq 0$.
B. $0 \leq \theta_0 < 1$.
C. $-2 < \theta_0 \leq 0$.
D. $0 \leq \theta_0 < 2$.
E. None of the above.

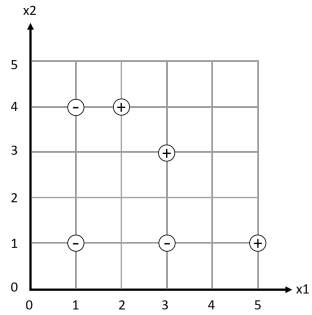
First write out the 4 inequalities corresponding to the 4 inputs. For example, for $x_1 \;=\; 1$and $x_2 \;=\; 1$, the inequality is $0.5(1) \;+\; 0.5(1) \;+\; \theta_0 \;>\; 0$. Then, solve the 4 inequalities to obtain the bounds for θ_0.

Q13-14: Say we apply perceptron learning to some binary classified data points as follows. Assume that the points are examined in the order given below; *i.e.* the negative points are observed first, then the positive points.

Instances classified as –1 are:
1. (1,1),
2. (3,1),
3. (1,4);

Instances classified as +1 are:
4. (2,4),
5. (3,3),
6. (5,1).

Pictorially:



Q13 (4 points): If the initial weight vector $\langle\theta_0, \theta_1, \theta_2\rangle$ is ⟨-15, 8, 3⟩ and the learning rate is 0.5. Which of the following is true of the decision boundary obtained after one epoch (*i.e.*, all points iterated through one time)? Mark one choice.

A. The line linearly separates the data.
B. The line does not separate all of the data correctly, but it can do so if the algorithm is run for more iterations.
C. The line does not separate all of the data correctly, and it cannot do so even if the algorithm is run for more iterations.
D. None of the above.

x_0	x_1	x_2	Misclassified?	Weight after
1	1	1	$sign(-4) == -1$ => No	⟨-15, 8, 3⟩
1	3	1	$sign(12) != -1$ => Yes	⟨-15.5, 6.5, 2.5⟩
1	1	4	$sign(1) != -1$ => Yes	⟨-16, 6, 0.5⟩
1	2	4	$sign(-2) != 1$ => Yes	⟨-15.5, 7, 2.5⟩
1	3	3	$sign(13) == 1$ => No	⟨-15.5, 7, 2.5⟩
1	5	1	$sign(22) == 1$ => No	⟨-15.5, 7, 2.5⟩

Thus the weight vector at the end of the epoch is ⟨-15.5, 7, 2.5⟩. This still misclassifies instances 2 and 3. However, since the points are linearly separable, the line will eventually be able to separate the data correctly (after 4 epochs).

Q14: Reconsider the same examples as above, but when the initial weight vector is ⟨0, 0, 0⟩ and the learning rate is 1. After running PLA for *n* iterations, we find that point #6 (5,1) is misclassified *p* times and point #3 (1,4) is misclassified *q* times. What are the final generated weights? Mark one choice.

A. $np\,\langle 1, 5, 1\rangle - nq\,\langle 1, 1, 4\rangle$.
B. $n\,\langle 1, 5, 1\rangle + n\,\langle 1, 1, 4\rangle$.
C. $p\,\langle 1, 5, 1\rangle - q\,\langle 1, 1, 4\rangle$.
D. $-p\,\langle 1, 5, 1\rangle + q\,\langle 1, 1, 4\rangle$.
E. $-p\,\langle 0, 5, 1\rangle + q\,\langle 0, 1, 4\rangle$.

Q15: You run gradient descent for 200 iterations with $\alpha = 0.2$ and compute $J(\theta)$ after each iteration. You find that the value of $J(\theta)$ decreases slowly and is still decreasing. Based on this, which of the following conclusions seems most plausible? Mark one choice.

A. Rather than use the current value of α, it would be more promising to try a smaller value of α (say $\alpha = 0.1$).
B. Rather than use the current value of α, it would be more promising to try a larger value of α (say $\alpha = 0.4$).
C. $\alpha = 0.2$ is an effective choice of learning rate.
D. Either a larger or a smaller value of α is promising.
E. We should stop iterating, as the parameters have converged.

Q16 (<u>6 points</u>): Recall that gradient descent changes the parameters along the negative of the gradient. Assuming a vector of parameters `x` and the calculated gradient `dx`, the weight update can be implemented using the following form in Python:

```
x += - learning_rate * dx.        # plain GD update
```

where `learning_rate` is a hyperparameter, some fixed constant.

Momentum update is another approach that almost always enjoys better convergence. This update can be motivated from a physical perspective. Initializing the parameters with random numbers is equivalent to setting a particle with zero initial velocity at some location. The optimization process can then be seen as equivalent to the process of simulating the weight vector (i.e. a particle) as rolling on the loss landscape. Note that this is different from the plain update shown above, where the gradient directly integrates the position. Instead, in momentum updates the gradient only directly influences the velocity, which in turn affects position.

If momentum is modelled with a variable `momentum`, a floating point fraction $\in \{0,1\}$, where a larger value indicates a greater mass, which choice implements correctly momentum update to replace the `????` in Line 2 below? Mark one choice.

```
# Momentum update
????        # L2. integrate velocity
x += v      # L3. integrate position
```

A. `v = momentum * v - learning_rate * dx`
B. `v = momentum * v + learning_rate * dx`
C. `v += momentum + learning_rate * dx`
D. `v = momentum * v - dx`
E. None of the above.

The correct answer is best understood as two halves. `momentum * v` denotes the amount of momentum in the system; a setting of 0 for `momentum` means no momentum, and 1 means no friction. The second half is the normal update as in the standard gradient descent (as above); note that the gradient points in the positive direction so we have to take the opposite direction to minimise the function (loss function).

Q17: In lecture where we discuss gradient descent, we discussed the shift from a step size η to a learning rate α, which is sensitive to the magnitude of the gradient. A potential problem with learning rate is that it is uniform for all *n* attributes. What happens if we know that we should increase the learning rate for one attribute but decrease for another? Does this makes sense and if so, what do we need to do to adapt our learning set up for it? Mark one choice.

A. Yes, it does. The switch from η to α in the lecture already caters for this.

B. Yes, it does. We would need to introduce per attribute α_ns to cater for this.
C. No, it does not. The learning rate should uniformly affect, as it does (correctly) with α already.
D. It may. It will depend on the particular loss function used with gradient descent.
E. None of the above.

This is the gist behind the Adaptive Gradient (Adagrad) optimizer. In essence, different dimension may need different learning rates. this increases the learning rate for sparser parameters and decreases the learning rate for ones that are less sparse. This strategy often improves convergence performance over standard stochastic gradient descent in settings where data is sparse and sparse parameters are more informative.

Q18-20: Bias and Variance

Q18: Which of the following are true about noise and error when using mean squared error as the loss function? Mark all responses that apply. If none fit, do not mark any.

A. We can reduce the effect of stochastic noise by increasing the complexity of hypotheses.
B. We can reduce the effect of stochastic noise by increasing the number of training data points.
C. We can reduce bias by increasing the complexity of hypotheses.
D. The model tends to overfit more easily as the level of noise grows higher.
E. The model tends to overfit more easily as the complexity of $f(x)$ increases.

Option C's "bias" can be interpreted as easier model bias or bias error, which gives opposing correct answers. We allow it to be either true of false. Option B is also both true and false depending on interpretation. We again allow Option B to be either to be true or false.

Q19: Which of the following are true about the hypothesis set, H, and average hypothesis, $\underline{h}$? Mark all responses that apply. If none fit, do not mark any.

A. The average hypothesis $\underline{h}$ only depends on H.
B. The average hypothesis $\underline{h}$ always lies in set H.
C. For binary classification, $\underline{h}$ must be a binary function.
D. Let H_2 and H_{10} be the 2nd and 10th order hypothesis sets, respectively. Then $H_2 \subset H_{10}$.
E. None of the above.

Q20: When a model is underfits, which of the following is likely to be the case? Mark all responses that apply. If none fit, do not mark any.

A. It has low bias error and high variance error.
B. It has high bias error and high variance error.
C. It has high bias error and low variance error.
D. It fits the training data well and fits the testing data poorly.
E. The model is a simple representation of a more complex reality.

Q21 (**4 points**): Let **H** be the hat matrix (*i.e.*, $H = X(X^\top X)^{-1}X^\top$) and I be an identity matrix of the same rank as **H**. In tutorial, we showed that $(H-I)^2 = H^2 - 2HI + I^2 = H - 2H + I = I - H$. What will be the value of $(H-I)^k$? Mark all responses that apply. If none fit, do not mark any.

A. $I - H$.
B. $(-1)^{k+1}(H-I)$.
C. $(-1)^{k+1}(I-H)$.
D. $(-1)^k(H-I)$.
E. $(-1)^k(I-H)$.

For $k = 2$,

$$(H - I)^2 = H^2 - 2HI + I^2 = H - 2H + I = (I - H) = (-1)^2(I - H)$$

For $k = 3$,

$$(H - I)^3 = H^3 - 3H^2I + 3I^2H - I^3 = H - 3H + 3H - I = H - I = (-1)^3(I - H)$$

Therefore, $(H - I)^k = (-1)^k(I - H) = (-1)^{k+1}(H - I)$.

Q22: According to Domingos' bias–variance decomposition framework discussed in tutorial, we have the following: $E_{D,y}[L(y, \hat{y})] = c_1 N(x) + B(x) + c_2 V(x)$. For 0–1 loss, if c_1 and c_2 are both negative, which of the following is true? Mark all responses that apply. If none fit, do not mark any.

A. Increasing bias will increase loss.
B. Increasing variance will increase loss.
C. Increasing bias will decrease loss.
D. Increasing variance will decrease loss.
E. Increasing noise will decrease loss.